“TESDA-Bot: An Interactive Guide for TechVoc Students to prepare for National Certificate (NCII) Exams”

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ESTOLE, JOHN ALBERT
ESTUYA, LAUCHE MARIE
PARINGIT, GEL
REFUGIO, DEMARCK JV
VOCAL, RICOMEL

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INTRODUCTION

Background and Rationale of the Study

Technical Vocational Education and Training (TVET) has gained increasing recognition as a critical pathway for workforce development in the Philippines, with the Technical Education and Skills Development Authority (TESDA) leading efforts through its National Certificate Level II (NCII) assessments. These certifications validate essential competencies in high-demand trades such as computer systems servicing, welding, automotive servicing, and household services, enabling graduates to secure immediate employment and contribute to economic growth. Globally, competency-based vocational training models demonstrate that targeted preparation significantly improves certification success rates and labor market alignment.

However, despite TESDA's extensive network of training centers and online resources like the Training for Work Scholarship Program (TWSP) and e-Learning platform (TOP), TechVoc students continue to face substantial challenges in NCII exam preparation. Existing methods—primarily static printed reviewers, classroom-based drills, and generic online modules—lack interactivity, personalization, and real-time feedback, resulting in persistently low pass rates that hover below 60% in many sectors. These limitations are particularly acute for students from rural areas and low-income households, who face financial barriers to costly review centers and limited access to qualified instructors, exacerbating the digital divide in skills development.

In the Philippine context, TESDA reports consistently show significant certification reluctance among NCII completers, with substantial attrition rates between training completion and examination. The Department of Labor and Employment (DOLE) has emphasized the urgency of innovative solutions to address skills mismatches, aligning with Republic Act 11230 (TVET Act) mandates for universal access to quality technical education that meets industry demands.

Given these challenges, the development of TESDA-Bot: An Interactive Guide for TechVoc Students to prepare for National Certificate (NCII) Exams aims to transform NCII preparation by providing a free, 24/7 AI-powered chatbot. This platform simplifies complex TESDA training regulations through conversational dialogues, delivers unlimited practice exams with hint-based feedback, supports self-paced learning, and creates a pressure-free environment to build student confidence and competence.

This study aims to solve the main research gap which exists because there are no intelligent tutoring systems that provide accessible learning resources for TESDA NCII preparation in the Philippines. The existing e-learning platforms and generic chatbots lack proper implementation of adaptive assessment simulation combined with progressive feedback mechanisms and TESDA-specific competency mapping to meet the special requirements of vocational certification examinations.

This study has major significance because it intends to make NCII certification results accessible to TechVoc students who currently lack educational resources. This study also aligns with SDG 4: Quality Education by promoting inclusive and equitable access to high-quality technical training. Ultimately, TESDA-Bot seeks to minimize certification costs, reduce system dropout rates, and empower students to master industry-ready competencies, fostering a more qualified and competitive Philippine workforce.

The expected outcomes of the project include three main benefits which will decrease certification expenses, reduce system dropout rates, and create secure learning spaces which will help students succeed while developing a vocational training system that uses AI technology to produce more qualified employees for the Philippine workforce.

Objectives of the Study

The main objective of this research is to build TESDA-Bot, an interactive AI assistant that is accessible 24/7 at your most convenient time without any pressure, offering free mentoring services to ensure students' success in acquiring NCII Certification.

To achieve this goal, the research pursues five specific objectives:

1. To provide students with opportunities to take control of their own review schedule by giving them ways to gauge how well they are doing and how far they have progressed.

2. To develop a safe environment where the students will be able to ask and practice the exams as many times as they need until they feel confident enough to take the actual NCII test on their own without any pressure.

3. To transform and intricate TESDA training rules into a much manageable, straightforward interactive dialogues that can effectively assist students in grasping information with the help of AI.

4.To offer an accessible guide that students can utilize via web app, ensuring that those who cannot afford expensive study resources or individuals who still lack access to a reliable reviewer can utilize it.

5.To establish a hint-based feedback mechanism where the chatbot not only provides the correct answer but also directs the student towards the proper reasoning during practice tests.

Significance of the Study

The development and application of TESDA-Bot represent a fundamental shift from conventional and static methods of vocational assessment to a more fluid and AI-driven model of education. The research is significant because it tackles head-on the digital divide and economic barriers that prevent Filipino TechVoc students from attaining professional certification

This study centered on the development and implementation of "TESDA-Bot: An Interactive Guide for TechVoc Students to Prepare for National Certificate (NCII) Exams" holds profound significance for various stakeholders within the Technical Vocational Education and Training (TVET) sector in the Philippines.

The following sector is the beneficiaries of this study:

Field of Technical Vocational Education and Training (TVET)

TESDA-Bot represents a novel approach to enhancing the preparation of Technical Vocational (TechVoc) students for their National Certificate (NCII) examinations. It contributes to the field by offering an innovative, accessible, and interactive learning tool that complements traditional review methods. This study aims to demonstrate how AI-powered chatbots can effectively bridge the gap in exam preparation resources, thereby improving the overall quality and accessibility of TVET education.

TechVoc Students

The primary beneficiaries of TESDA-Bot are the TechVoc students themselves. The interactive guide will provide them with a readily available, personalized platform to review course materials, practice exam questions, and receive immediate feedback. This will not only boost their confidence but also improve their understanding of the NCII exam requirements, ultimately increasing their chances of passing and obtaining their certifications. This directly addresses the problem of students facing challenges in accessing adequate and tailored review materials.

TESDA and Training Institutions

For TESDA and various training institutions, TESDA-Bot offers a scalable and cost-effective solution to support student success. It can serve as a valuable tool to augment their existing review programs, ensuring that students are better prepared for assessments. The bot's ability to provide consistent and up-to-date information can also help maintain high standards across different training centers.

Educators and Trainers:

Educators and trainers can utilize TESDA-Bot as a supplementary resource to reinforce learning and provide additional support to students outside of regular class hours. It can help identify areas where students commonly struggle, allowing educators to focus their instruction more effectively and provide targeted interventions.

To Policymakers and Future Researchers

This study's findings can inform policymakers about the potential of AI in enhancing vocational education and skills development. It provides evidence for the effectiveness of digital learning tools in preparing students for national competency assessments. Furthermore, the research can serve as a foundation for future studies exploring the integration of AI in various aspects of vocational training and assessment, contributing to the evolving body of literature in educational technology and workforce development.

Scope and Limitation

This study aimed to develop and evaluate TESDA-Bot, an interactive AI guide designed to assist TESDA TechVoc students at selected TechVoc schools in

Metro Manila in preparing for their NCII exams, covering key competency areas in qualifications like NCII in Computer Systems Servicing and Bread and Pastry Production. The study spanned within the duration of the school year 2025-2026, targeting a selected group of these students for testing feedback on its features, including review materials, practice quizzes, FAQs, and exam strategies. However, it is important to note limitations such as coverage of only popular NCII qualifications, a small sample size affecting generalizability, reliance on participants' device/internet access and digital literacy, shallower content depth as a supplementary tool, AI constraints on nuanced responses, challenges in measuring exam impact due to data policies, and finite resources for development and maintenance. These delimitations ensured focused, feasible implementation while prioritizing practicality and resource constraints. Despite them, TESDA-Bot holds potential to enhance NCII exam preparation through user-friendly, accessible support.

Conceptual Framework

INPUT		PROCESS		OUTPUT
➢ *Student Data (Learning Needs, Prior Knowledge)* ➢ *NCII Examination Content (TR's, Sample Qs)* ➢ *Technological Infrastructure* ➢ *Expert Knowledge*	→	➢ *User Interaction* ➢ *NLU & Query Processing* ➢ *Content Retrieval & Response Generation* ➢ *Quiz Module* ➢ *Learning Support* ➢ *Feedback Mechanism*	→	➢ *Enhanced NCII Exam Preparedness* ➢ *Increased Confidence* ➢ *Accessible Review Materials* ➢ *TESDA-Bot System (Functional Chatbot)* ➢ *Contribution to TVET (Improved Pass Rates, Data for Improvement)*

Figure 1. TESDA-Bot Conceptual Framework

This study employs an Input-Process-Output (IPO) model as its conceptual framework to delineate the relationships between the core components and expected outcomes of developing and implementing TESDA-Bot.

Input:

This section details the essential resources, data, and foundational elements required for the creation and operation of TESDA-Bot.

1. **Student Data:**

○ **Demographic Information:** Grade level, course/specialization, school.

○ **Learning Needs:** Identified areas of difficulty or topics students find challenging in NCII preparation.

○ **Prior Knowledge Assessment:** Results from initial diagnostic tests or self-assessments of their understanding of NCII competencies.

○ **Learning Preferences:** Preferred modes of learning (e.g., text-based, visual aids, practice questions).

2. **NCII Examination Content:**

○ **TESDA Training Regulations (TRs):** Official documents outlining the competencies, knowledge, skills, and attitudes required for specific NCII qualifications.

○ **Sample Examination Questions:** Past exam papers or realistically crafted practice questions aligned with TRs.

○ **Key Concepts and Procedures:** Essential theoretical knowledge and practical skills for each competency unit.

○ **Assessment Criteria:** Understanding of how NCII exams are evaluated.

3. **Technological Infrastructure:**

○ **Chatbot Development Platform:** The chosen software or framework for building the conversational AI (e.g., Dialogflow, Rasa, custom development).

○ **Natural Language Processing (NLP) Capabilities:** Algorithms and models for understanding user queries and generating relevant responses.

○ **Database:** Storage for NCII content, user interaction logs, and performance data.

○ **User Interface (UI) / User Experience (UX) Design:** Principles for creating an accessible and intuitive chatbot interface.

4. **Expert Knowledge:**

○ **TESDA Trainers/Assessors:** Input on critical competencies, common student errors, and effective review strategies.

○ **Curriculum Developers:** Guidance on structuring the learning content logically.

○ **AI/Chatbot Developers:** Technical expertise in chatbot design and implementation.

Process:

This component describes the operational flow and interactions within the TESDA-Bot system, detailing how inputs are transformed into outputs.

1. **User Interaction:**

○ **Query Input:** Students interact with TESDA-Bot by asking questions, requesting information, or initiating practice quizzes.

○ **Natural Language Understanding (NLU):** The bot processes user queries using NLP to understand intent and extract key information.

2. **Content Retrieval and Response Generation:**

○ **Information Search:** The bot accesses its knowledge base (NCII content) to find relevant information or answers.

○ **Personalized Feedback:** Provides explanations, definitions, step-by-step procedures, and answers to specific questions.

○ **Quiz Module:** Generates practice quizzes based on selected competency areas, tracks user responses, and provides immediate feedback on correctness.

3. **Learning Support Features:**

○ **Progress Tracking (Basic):** Logs user interactions and quiz performance to provide a basic overview of areas covered or needing more review.

○ **Guidance on Exam Strategies:** Offers tips and advice on how to approach the NCII examination.

4. **Feedback Mechanism:**

○ **User Feedback Collection:** Allows students to rate the helpfulness of responses or report issues.

○ **Data Logging:** Records interaction patterns, common queries, and quiz results for analysis and improvement.

Output:

This section outlines the expected outcomes and results derived from the use of TESDA-Bot.

1. **Enhanced NCII Exam Preparedness:**

○ **Improved Knowledge Retention:** Students gain a better understanding and recall of NCII competencies.

○ **Increased Confidence:** Students feel more prepared and confident in their ability to pass the NCII exams.

○ **Skill Reinforcement:** Practice quizzes help reinforce practical skills and theoretical knowledge.

2. **Increased Accessibility to Review Materials:**

○ **24/7 Availability:** Students can access review support anytime, anywhere.

○ **Personalized Learning Pace:** Students can learn and practice at their own speed.

3. **TESDA-Bot System:**

○ **Functional Chatbot:** A deployed and operational interactive guide.

○ **User Interaction Data:** Logs of student queries, quiz performance, and feedback.

4. **Contribution to TVET:**

○ **Improved Pass Rates (Potential):** An anticipated increase in the success rate of TechVoc students in NCII examinations.

○ **Data for Improvement:** Insights gained from user data can inform future iterations of the bot and potentially TESDA's review strategies.

○ **Foundation for Future Research:** Provides a basis for further studies on AI in vocational education.

Definition of Terms

For the readers to fully comprehend this study, the following are the notable terms that have been utilized in the study:

Operational Term

● **Technical Vocational Education and Training (TVET):** Refers to the education process that equips individuals with the knowledge, skills, and competencies for employment in a particular trade, occupation, or profession. In the context of this study, it specifically pertains to the programs and certifications offered by TESDA.

● **National Certificate (NCII) Exams:** These are competency-based assessments administered by TESDA to validate that individuals possess the required skills and knowledge for specific technical and vocational roles. Passing these exams leads to the awarding of an NCII certificate.

● **TESDA-Bot:** An AI-powered chatbot developed in this study, designed to provide interactive guidance and review materials for TechVoc students preparing for their NCII exams. It functions as a 24/7 accessible digital assistant.

● **Chatbot:** A computer program designed to simulate conversation with human users, especially over the internet. In this study, it refers to the AI assistant that interacts with students.

● **AI-powered:** Utilizing Artificial Intelligence technologies, such as Natural Language Processing (NLP), to enable the chatbot to understand user queries and provide intelligent responses.

● **Self-Paced Learning:** A mode of learning where students control the pace at which they progress through the review materials and practice exams, allowing them to learn according to their individual needs and schedules.

- **Hint-based Feedback:** A type of feedback provided by the chatbot that not only indicates whether an answer is correct or incorrect but also guides the student towards the correct reasoning process.
- **Safe Environment:** Refers to the non-pressured, supportive digital space created by the chatbot where students can practice and ask questions without fear of judgment or negative consequences, thereby boosting their confidence.

Technical Term

- **Technical Vocational Education and Training (TVET):** (As defined operationally above, but also understood in the broader educational and policy context related to skills development and workforce readiness.)
- **National Certificate (NCII) Exams:** (As defined operationally above, but also understood within the framework of TESDA's national qualification system and industry standards.)
- **Artificial Intelligence (AI):** The simulation of human intelligence processes by computer systems, including learning, problem-solving, and decision-making.
- **Chatbot:** A computer program that simulates human conversation through text or voice interactions.
- **Natural Language Processing (NLP):** A subfield of AI that enables computers to understand, interpret, and generate human language.
- **Training Regulations (TRs):** Official documents issued by TESDA that detail the specific competencies, knowledge, skills, and attitudes required for a particular vocational qualification.
- **Competency-based assessment:** An evaluation method that focuses on determining whether an individual can perform specific tasks or demonstrate specific skills to a defined standard, rather than on theoretical knowledge alone.
- **Attrition Rate:** The rate at which students drop out of a program or fail to complete a course or assessment.

CHAPTER II

REVIEW OF RELATED LITERATURE

This chapter reviews literature and studies supporting technical-vocational education (TVET), middle-skills development, and AI-chatbot integration in educational and career technologies, organized into local literature/studies and foreign literature/studies to contextualize workforce linkages. Philippine TVET research reveals alignments with employment needs amid challenges like industry mismatches, inconsistent implementation, and competency gaps, with student success hinging on teacher skills, resources, support, and guidance—necessitating reforms and partnerships. Local studies affirm TVET/TESDA efficacy while highlighting barriers and innovations like gamified/AI tools for engagement and readiness. Foreign literature underscores AI chatbots' role in personalized learning, despite engagement, privacy, and integration hurdles, while studies demonstrate their enhancement of technical/vocational skills via interactive systems, tempered by tech/security/training issues, and amplified by gamification/microlearning.

Local Literature

The existing Philippine literature indicates that students who choose their own career paths fail to select fields which match their actual capabilities and their interest and their academic preparedness, which creates an urgent requirement to enhance student guidance systems. Public high school students in Cebu City base their career choices on what their families expect and what they see as popular trends, which leads to their career confusion and demotivational experience and their inability to predict their future outcomes[1]. The study proposes a career path recommender system which targets the problem of career choice mismatches that students in the Philippines experience. The tool helps students make important choices which determine their future career paths by connecting their personal interests and abilities with available professional options. The system uses machine learning together with a multi-dimensional dataset to create customized recommendations which help students select paths that match their individual academic and personal characteristics[2]. The research conducted in Tagbilaran City Division shows that National Career Assessment Examination results do not affect students' choice of academic tracks because students choose the Academic Track despite better performance in other tracks like Arts Track, which proves that career guidance programs fail to execute their functions[3]. The University of the Philippines Diliman study found that students who choose college majors that do not match their Senior High School strands face academic challenges in STEM fields which demonstrates the necessity of proper track alignment[4]. This study is about choosing the right career before entering college which functions as a vital turning point for students yet most students select their career paths without professional assistance and they fail to use essential resources such as NCAE tests and aptitude assessments. The absence of prior guidance results in two main problems for students because it creates academic performance problems and students select study paths which do not match their personal traits and interests and natural talents. The researchers call for a system development which will deliver precise guidance through the use of interactive tools that include image-based questionnaires to enhance student participation and focus according to their study[5]. The present research examines middle-skilled professions in the Philippines through examination of institutional definitions and execution methods for middle-skilled training programs which determine their success in meeting workforce needs outlined in the Philippine Development Plan (PDP) 2023-2028. The research employs a multi-method qualitative approach which combines data from the Philippine Statistics Authority's Labor Force Survey with interviews of TVET practitioners to demonstrate that Level 5 post-secondary graduates achieve better employment outcomes and higher salary levels than Level 6 bachelor's degree undergraduates. Training programs face effectiveness challenges because of two main issues which include a partial understanding of middle skills and an educational system that fails to match actual workforce requirements. The research study recommends development of a central policy framework which requires ongoing input from all affected parties to protect middle-skills education from becoming outdated through technological advancements. The nation requires complete reform measures which will enable its workforce to adapt while boosting its international competitiveness. Future research should evaluate how middle-skills training impacts employment results and economic resilience over an extended period of time. The nation requires complete reform measures which will enable its workforce to adapt while boosting its international competitiveness. Future research should evaluate how middle-skills training impacts employment results and economic resilience over an extended period of time [6]. The research paper presents results from a pilot study which tested the Hotel and Restaurant Services and Electro Mechanic Course at the University of San Carlos between 2012 and 2015. The university partnered with the Center for Industrial and Technical Enterprise to assist graduates in finding employment opportunities. The university provided general education subjects, while the center focused on technical training. The students gained practical experience through job training which resulted in their employment after graduation because they had developed essential skills for the workforce. The pilot study demonstrated that competency-based training successfully fulfills the requirements of academic institutions and industries [7]. The study examines how public secondary schools in Pagbilao District Quezon implement TLE and TVL programs which aim to create lifelong learners through their new curriculum development. The research used descriptive developmental research and sequential exploratory research to study both school administrators and TLE and TVL teachers. The study found that both programs operate at sufficient levels because they maintain all required components through their qualified teaching staff. The program showed moderate implementation while students achieved competent learning results. The study established that both TLE and TVL programs operate at acceptable levels which connect program implementation assessment with student performance results

[8]. The research examined work readiness among TVL graduates from two public secondary schools located in Mabalacat Pampanga through their assessment of social skills and personal attitude and organizational awareness and technical skills. The study assessed employed graduates from 2019-2022 who mostly consisted of male participants aged 22-24 and had limited work experience. Graduates demonstrated proficiency in required areas although their work readiness showed significant variation according to their different profiles. The assessments conducted by employers and graduates showed no significant differences between their evaluation results. The proposed intervention plan will improve work readiness for graduates through its two main components which include ongoing employability enhancement initiatives and the tracer study that will monitor employment status [9]. The study evaluated A competency-based program implementation at a public high school technical vocational education program through its four objectives which included 1) evaluation of program execution 2) identification of operational challenges 3) assessment of different perceptions between two groups about program execution and operational difficulties 4) development of program improvement recommendations. The researchers used descriptive research methods to collect data through a validated questionnaire which used the Five-Point Likert Scale. The researchers used statistical analysis methods which included frequency percentage weighted mean standard deviation and t-test analysis to test hypotheses at a significance level of 0.05 [10].The assessment of Senior High School student performance in the Strengthened Technical-Vocational Education Program (STVEP) at Isabela

School of Fisheries is conducted through this study for the academic year 2024-2025.The study used questionnaires to collect data from 126 students and teachers while investigating specializations that included Aquaculture, Carpentry, Food Processing, and Organic Agriculture Production. The research found that students who achieved high performance in Aquaculture studies and Organic Agriculture studies displayed three different levels of skill mastery. The results of the study depended on student interest, study habits, teacher characteristics, teaching methods, administrative support and available resources. Students and teachers showed significant perception differences about Aquaculture and Carpentry according to the research findings. The study recommends enhancing teacher training, instructional materials, school infrastructure, and parental engagement, emphasizing their roles in improving technical-vocational education outcomes and guiding future curriculum and development efforts [11]. The Senior High School program in the Philippines has highlighted technical vocational education (TVE) which helps students prepare for jobs that require middle-level skills. The study assessed how public senior high schools in Batangas operate their Technical Vocational Livelihood (TVL) Track through the evaluation of input resources and their adherence to Department of Education standards and their educational processes and student satisfaction levels. The research employed a descriptive-evaluative design which collected data from 36 school heads and 119 TVL teachers and 375 students through surveys and interviews and documentary analysis. The study found that organizations received enough resources to operate their programs while they achieved above-standard results and reached very high process implementation levels and students expressed strong satisfaction levels. The analysis established significant relationships between satisfaction and both input adequacy and compliance which showed different results according to participant evaluations. The study needs additional research about TechVoc graduates' work opportunities to determine the TVL track benefits [12]. The study explored factors that determine Technical Vocational Education (TVE) students' choice of specialization and their academic success at DepEd Division of Sorsogon. The study examined 405 Grade 10 students through quantitative-descriptive research which revealed that demographic elements and environmental elements together influenced students' specialization selection. Students achieved proficiency in Food Trades and Cosmetology and other fields while their performance in SMAW FCM and Civil Technology reached an approaching proficiency level. The results showed that academic success related to demographic characteristics but environmental factors did not show any significant connection to academic success. The study found that family income and job opportunities function as the main factors which drive TVE students to select their specialized fields of study. 13. The study found that family income and job opportunities function as the main factors which drive TVE students to select their specialized fields of study [13]. Students choose their vocational path during their entrance to senior high school while school counselors assist them with their vocational selection process. The Career Guidance Advocacy Program investigation studied its operational execution in Bohol which assessed its effect on 1200 students and 60 guidance counselors and homeroom advisers. The research used a descriptive correlational design with adapted questionnaires to show that educational qualifications and job descriptions together with work experience and resource limitations affected the success of career guidance programs. The National Career Assessment Examination (NCAE) results went unobserved by multiple students because of various intervening factors which demonstrated that career guidance had no direct impact on student decision-making. The program required improvement through recommendations which researchers presented [14]. The study examines how Technology and Livelihood Education teacher competencies affect student learning outcomes because this research seeks to identify academic success predictors. The study used a descriptive research design to collect data from TLE teachers and students in La Union, Philippines by using questionnaires that assessed teacher competencies and background information. The analysis showed that TLE teachers achieved high levels of instructional research and extension competence which resulted in positive student outcomes particularly in Home Economics. The research findings demonstrate that teachers' educational background directly influences their ability to teach and conduct research which subsequently determines student academic achievement. The research study establishes a foundation for improving TLE instruction through specific enhancement measures [15].

Local Studies

The literature review includes studies about Technical-Vocational Education and Training (TVET) and TESDA programs and educational technology implementation in Philippine educational systems. The research study describes TESDA scholarship data and employment statistics and stakeholder interviews from 2019 to 2024 as the basis for TVET evaluation which demonstrated that data-based policies and government-industry-training institution partnerships lead to better program results[16]. The research found that BSICC students achieved satisfactory results in contact center competencies because they met TESDA standards for communication and teamwork skills yet they needed to develop their technical operations expertise to secure international job opportunities[17]. The research on Technology and Livelihood Education (TLE) demonstrated that teacher qualifications and available facilities determine student performance yet equipment and resource shortages in Cookery and Horticulture training programs continue to impact student learning outcomes[18]. The National System of Technical Vocational Education and Training (NSTVET) review discovered that TESDA requires better regulatory operations and enterprise training approaches because the organization needs to develop programs which match industry needs while increasing its funding capabilities and monitoring systems and public relations efforts[19]. The study examined AI-driven English learning tools used in Philippine colleges and found that these tools enable students to learn independently while delivering personalized teaching methods and fostering educational innovation. The study found three primary obstacles which must be addressed before educational AI systems can reach their maximum potential. The three obstacles include scarcity of training materials and infrastructure problems and ethical issues which need to be fixed before educational AI systems can function at their best capacity[20]. The study addresses the three main problems which affect TESDA-accredited technical-vocational education programs because all three exist through their need to use heavy and costly equipment tools. The Notable Ergonomic Universal Working Table was created through research development which followed the ADDIE model for evaluation by 30 participants who included TESDA assessors and students. The table received high ratings for functionality, efficiency, usability, and maintainability, with overall effectiveness rated highly as a multi-functional and portable instructional asset. The solution demonstrated a teaching enhancement which led to better educational outcomes while satisfying TESDA requirements. The research demonstrates that the table enables better teaching methods and the study suggests that future designs should include smart technologies to create systems which meet the requirements of Industry 4.0 and Education 5.0 [21]. CSS Quest is an educational initiative that helps Grade 12 students in Calamba City, Laguna learn about computer system servicing. The program creates an interactive learning environment through its gamified progress tracking system and interactive storytelling algorithm. Students can monitor their progress through the tasks and challenges they complete, while they gain rewards and experience content through narrative-driven scenarios that simulate real-life situations. The approach helps students develop critical thinking skills and knowledge retention through active learning, which helps them prepare for real-world professional challenges [22]. The study uses a survey questionnaire to evaluate the Technology and Livelihood Education competencies which Grade VI teachers possess. The study results show that most TLE teachers belong to the age group of young adults to middle-aged adults while they are married and work at Teacher I positions and most of them have completed master's degree coursework. The group of teachers who work at the school system needs to obtain national certification and complete training programs because they currently lack both these requirements. The profile of TLE teachers varies according to their age and gender and marital status and position title and educational level and training they have completed. The research indicates that TLE teachers possess "experienced" status because they demonstrate proficiency in most competencies which exist across four TLE areas while their ICT and Entrepreneurship and Agriculture and Home Economics and Industrial Arts competencies show no connection to their teacher profiles except for Agriculture which displays a connection based on position title [23]. The research examined factors that affect STEM teaching at a Private Higher Education Institution in Maasin City during the 2017-2018 academic year. The researchers used descriptive research methods to evaluate teacher abilities and school environment and educational resources which students used to study STEM subjects. The study discovered three main results: (a)Most STEM teachers were young married women who had completed master's level education but required additional K to 12 training. (b)The teaching staff showed high proficiency in their instructional abilities while the school environment and educational resources received proficient ratings. (c)Most students showed very satisfactory results while they developed positive attitudes toward STEM subjects. (d)The researchers discovered a relationship between K to 12 training hours and teacher proficiency and they also found a relationship between curriculum content and pedagogy and bachelor degree courses while school climate showed a link with gender. (e)The study found no important connection between K to 12 implementation factors and student academic performance. (f)Teachers demonstrate their dedication to effective educational practices through their implementation of best practices in their teaching [24]. The study evaluated the implementation of the Technical-Vocational-Livelihood (TVL) Track in secondary schools in the Botolan District. The research team studied six school heads together with 75 teachers through a survey questionnaire which they analyzed using descriptive research methods and both descriptive and inferential statistics. The TVL track showed successful implementation through its facilities and industry partnerships and learning environments and instruction and overall requirements of the program although teachers and administrators showed different views about its execution. The study highlights the need to assess responses for Senior High School TVL Track evaluation while recommending future research with larger participant groups from multiple areas to strengthen study findings [25]. The study investigated how high school graduates from model schools in CALABARZON region experienced their transition to life after graduation. The study involved 80 graduates who completed their studies at three different Technical-Vocational schools. The research team collected data through interviews which they evaluated using a qualitative interpretative phenomenological research design. The study found that students' transition process was affected by five major factors which included their perception of others and school programs and their senior high school difficulties and their post-graduation career decisions and their self-preparation methods and track selection decisions. The study recommends that institutions should develop career guidance programs which will assist both students and their parents to identify suitable extracurricular activities while developing systems to acknowledge student accomplishments and establish a graduate tracer study program [26]. The Enhanced Basic Education Act of 2013 aimed to improve educational standards in the Philippines, which permitted students to exit study programs through four different paths that included employment opportunities, entrepreneurship, tertiary education, and middle skills development. A Descriptive Correlation Research conducted with 564 Senior High School (SHS) graduates in Plaridel, Bulacan, showed that more than 75 percent of students continued their education at tertiary levels. A small percentage opted for employment (10.11%) and entrepreneurship (7.09%). The study discovered that SHS track selection by students resulted in their specific curriculum pathway choices which confirmed the effectiveness of Career Guidance and Counseling Programs. The researchers proposed extra activities which would help graduates make better career choices [27]. The study investigates the reasons and difficulties which twelve BTLEd students face who study Technology and Livelihood Education during the pandemic. The research used a phenomenological methodology together with thematic analysis to identify four themes which included academic progression financial difficulties and challenges in obtaining additional learning resources and the difficult nature of TLE as an academic discipline. The research findings show that students needed face-to-face interaction to develop their essential skills. The study suggests that future research could further explore specific dimensions of the TLE program [28]. The study investigates how NC2 certification helps improve job opportunities for workers in Cagayan de Oro City's tourism industry. The study investigates how staffing and job placement and promotion and job security and certification value and benefits affect their work performance. Research questions investigate three main areas which include the tourist industry demographic data of respondents and the TESDA Certificate employment assessment and the employment differences between various tourism certification holders. The research used a quantitative design which included surveys that targeted NC2-certified professionals and tourism industry employers [29]. Successful workplace achievements require proper communication skills which Senior High School students need to acquire their Technical-Vocational-Livelihood competencies. The study investigates SHS-TVL students' communication methods and their practical work abilities through self-assessments and teacher evaluations which include data from 330 students and 108 teachers. The questionnaire assessed communication skills through five categories which included verbal communication, written communication, non-verbal communication, active listening, and problem-solving abilities while measuring work readiness indicators. The results show that students need to improve their communication skills because teachers found their non-verbal and written communication abilities to be below average. The research establishes a strong positive correlation between communication skills and work readiness, which identifies non-verbal communication as the top predictor of employment readiness through its statistical relationship ($\beta = 0.27$, $p < 0.001$). The research demonstrates that dedicated communication training programs need to become part of the TVL curriculum because they require role-playing exercises and business partnerships to improve SHS-TVL graduates' communication skills and employment opportunities [30].

Foreign Literature

The literature review shows that educational areas now use chatbot technology and artificial intelligence (AI) systems to improve their learning processes. Through their ability to create customized educational materials and provide answers to questions and support administrative tasks, chatbots function as virtual assistants who deliver educational assistance to both teachers and students. The human role in teaching which includes emotional support and character development cannot be replaced by chatbots according to [31]. The research on second language acquisition shows that AI-driven chatbots help students develop language abilities through their personalized practice system and immediate feedback mechanism, but the technology needs to advance its capacity to handle complex dialogues and emotional situations that require ongoing research and human guidance according to [32]. Nursing education programs use chatbot-based systems to help students learn independently while they develop their clinical reasoning capabilities and their ability to operate specialized equipment such as mechanical ventilation according to [33]. AI tools which include chatbots enhance student engagement in secondary education while offering instant assessment capabilities, but schools must resolve three main challenges which include restricted technology access, insufficient teacher training, and potential negative effects on students critical thinking abilities according to [34]. The studies on tourism education show that universities need to create AI competency standards which should function as core curriculum requirements because these standards will boost students employability while developing educational programs that fulfill global digital market needs through faculty development and business collaborations according to [35]. The research shows that chatbots deliver several benefits to users because they establish custom-built experiences and provide continuous access to their services throughout the entire day. People frequently use these systems as personal assistants and shopping tools, but researchers have only begun to investigate their academic applications which support organizational activities during studies and courses and exams. The research creates an educational chatbot system which helps students and professionals through John Holland's framework and RIASEC assessment to identify their primary career-related personality traits [36]. The study demonstrates that universities need to deliver top-notch consultation and information services which will help students achieve satisfaction during their admission process. The research introduces Dinus Intelligent Assistance (DINA) as a solution which uses knowledge-based machine learning technology to create a conversational agent that automatically handles student inquiries. The system was developed using data from a university guest book containing common admission questions and achieved a high success rate in answering sample queries correctly [37]. The research investigates how educational technology has developed new teaching methods through its quick growth which enables teachers to use virtual assistants and chatbots for better student learning outcomes. The study examines current educational chatbot developments while identifying critical challenges that affect their design process and their implementation in e-Learning systems. The authors present a framework which shows all critical factors that need to be addressed for successful chatbot implementation [38]. In this study, the advancement of AI technologies in the classroom requires a stronger connection to pedagogical perspectives, particularly concerning student motivation. The research used self-determination theory to study how teacher support impacts student expertise, which includes digital literacy and self-regulated learning, on their intrinsic motivation to use chatbots. The study found that chatbots help students learn better, but students need teacher help to use them properly because this assistance fulfills their need to connect with others [39]. The use of chatbots has grown rapidly across various fields, including healthcare, marketing, and education. The research work develops an architectural design together with a communication framework, which delivers precise information to students. The system implements natural language processing techniques together with domain ontologies to identify specific questions and deliver appropriate responses [40]. The research shows that chatbots now serve as the main method for delivering intelligent support which enables faster assistance through their ability to interpret user inquiries and provide immediate correct responses. The research presents a prototype developed for the educational domain specifically designed to support university students in their coursework. The system architecture uses natural language processing (NLP) and domain-specific ontologies to control communication while it identifies user inquiries for accurate response delivery [41]. The conversation practice is essential for language learners but remains difficult and expensive to access. The existing gap in language practice requires mobile chatbots as a practical solution, yet these chatbots have not yet developed into effective practice tools. The study showed that a student's previous language skills together with their experience of learning more through chatbot interactions which included communication challenges serve as two main factors that determine student interest and engagement in this study [42]. The educators can use chatbot technology and conversational agents for teaching and learning purposes because AI platforms have developed Google Dialogflow which enables users to design conversations without needing advanced programming skills. The research examines a case study where a rule-based chatbot was integrated into both online and face-to-face training environments to augment and address specific challenges encountered in teaching [43]. The chatbots serve as real-world demonstrations of new artificial intelligence technology which uses research-based algorithms to improve efficiency and communication and dialogue capabilities. The tools enable users to create human-like conversations by using either audio or text while the system operates through its ability to recognize specific words and make choices. Students can experience a guided yet interactive learning experience through chatbots which use basic flowchart systems to control their conversation management [44]. The rapid growth of AI and chatbots in education and research brings educational institutions three transformative opportunities which they must handle with their associated ethical challenges. The research demonstrates that educational institutions need to develop fresh assessment methods and research methodologies because technological improvements in their assessment systems and research processes require them to adapt through creative solutions instead of treating AI as a danger. Key considerations include the role of AI in supporting human judgment, the potential for misuse, and the necessity of establishing solid ethical values and legislation to protect educational systems [45].

Foreign Studies

The existing research demonstrates that educational systems now use artificial intelligence systems, which include chatbots, to improve their teaching and learning processes and vocational training programs across various educational settings. The Philippine educational system conducted a qualitative study which found that AI chatbots assist pre-service Physical Education interns in developing essential teaching skills through their support of instructional planning and content organization and professional communication tasks which lead to increased confidence and understanding of concepts during practical work experience[46]. The Vietnamese tourism education research shows that for AI-related competencies to meet global standards and enhance graduate employability in an automated sector, educational institutions must improve their curricula and train staff in digital skills while establishing partnerships with industry companies[47]. The research review on AI applications in technical skills training shows that AI enables personalized learning and enhances vocational education systems in Industry 4.0 environments, but successful implementation requires organizations to solve data privacy issues and teacher training deficiencies and access disparities[48]. The descriptive study about Technology and Livelihood Education (TLE) shows that equipment and facility restrictions impede student achievement in practical subjects like cooking and horticulture despite teachers holding proper qualifications and receiving extensive training[49]. The research study on gamified chatbot usage in vocational certification programs shows that the combination of AI and gamification leads to higher student participation and satisfaction and improved learning results than conventional methods and non-gamified systems which make it an effective teaching method that can be expanded to wider use[50]. According to this research study a chatbot functions as a sophisticated software application which enables people to communicate with their digital devices while its main purpose is to deliver answers to student questions. The research shows that current chatbots develop security vulnerabilities which enable hackers to gain unauthorized access to information that passes through their systems. The study implements encryption as a security measure to stop unauthorized access to data during its transfer process and uses authentication session timeouts to limit user access time to the system [51]. The study defines a chatbot as a software program that creates human-like dialogue through Artificial Intelligence technology. The research presents the development of a student chatbot integrated with a website, utilizing Recurrent Neural Networks (RNN) for language processing and Convolutional Neural Networks (CNN) for image recognition. The system uses Dialogflow to establish intent representation and keyword matching which creates an interface that is both balanced and easy-to-use for all target users [52]. The research study used quasi-experimental methods to examine how a chatbot-based micro-learning system affected student learning motivation and academic performance. The results show that students who use the chatbot system demonstrate learning capabilities that match those of traditional classroom instruction. The study discovered that students who used the chatbot exhibited increased intrinsic motivation because they achieved their academic work through two main factors which involved their ability to make activity choices and their assessment of activity importance [53]. The research in question concerned how student perception and their assessment of a given unit varied when AI-driven chatbots were used. The created chatbot system used Dialogflow software together with Telegram instant messaging to deliver video content which helped students with their visual and auditory learning needs. The study results demonstrated that there was no academic performance difference between the two groups yet students who used the chatbot system experienced improved results in their online educational activities. The participants found the tool to be both enjoyable and helpful which enabled them to study and practice their learning material beyond classroom time [54]. The study demonstrates that Goal-Oriented Requirements Engineering methods which include the KAOS framework successfully enable stakeholder communication yet present implementation difficulties for beginner engineers. The research introduces KAOSbot, an AI-powered chatbot that uses Natural Language Processing (NLP) to act as a modeling assistant for these professionals. The study results showed that participants considered the tool straightforward to use while they found it beneficial, and they expressed strong intent to use the tool for requirements modeling in future projects [55]. The research demonstrates that using the "Einstein" chatbot system for physics instruction through technological platforms helps students who struggle to learn without teacher assistance. The study used the Multimedia Development Life Cycle (MDLC) framework which includes five stages: analysis, design, development, implementation and assessment to develop a mobile-based solution that meets practical requirements. The chatbot technology provides students with advanced communication support because smartphones serve as their main communication device which allows them to learn beyond the classroom [56]. This study, chatbots demonstrate significant potential as language learning tools, specifically for improving vocabulary acquisition. The research showed that chatbots enhanced student performance in both classroom-based and one-on-one tutoring environments. The analysis used Technology Acceptance Model (TAM) to show that perceived usefulness serves as the main factor which predicts students' tool usage intentions [57]. The research uses a web-based chatbot as a natural language dialogue system to help foreign language students who cannot practice with native speakers. The study found that students maintained interest in the system because they could access it at all times and they felt more confident using the non-judgmental computer program [58]. This research shows that students face difficulties when choosing their elective courses because they must handle academic interests and their actual graduation requirements and course availability and course level difficulties. The research presents EASElective as a new solution which works with existing advising methods and peer conversations to provide students with a natural language interface for their studies. The system manages a wide range of topics, from official course data to informal student opinions, utilizing components like intent detection and dialogue management to streamline the decision-making process [59]. This study, higher education institutions are increasingly utilizing chatbots as virtual assistants to help prospective students access information through natural language simulations. The research proposes a chatbot using a Retrieval Augmented Generation (RAG) approach, which combines cosine similarity search for document retrieval with an LLM for answer generation. The system enables users to enhance context understanding through its capability to store and retrieve previous chat interactions. Evaluation using metrics such as Recall, Precision, BLEU, and ROUGE scores confirms the effectiveness of the retrieval and generation modules [60].

Synthesis

The reviewed local and foreign literature and studies demonstrate that technical-vocational education together with emerging technology implementation serve as essential components which help students develop skills needed for employment and achieve career readiness. The educational systems which the researchers examined across all sources share a common goal to provide students with essential practical skills and industry-required competencies and knowledge. The initiatives face three major challenges which include skills mismatch problems and insufficient resources and gaps between program implementation and actual needs.

Local studies and literature show a common research focus which shows Technical-Vocational Education and Training (TVET) and middle-skills development as critical factors that improve job prospects for people. The research demonstrates that competency-based training together with industry partnerships and hands-on learning experiences all contribute to better work readiness and employment outcomes for students. Student performance depends on teacher competence and facility availability and curriculum design and institutional support according to research findings. The research shows that local studies demonstrate ongoing problems which include schools with limited resources and inadequate facilities and educational programs that do not meet actual job market demands. The research shows that career guidance programs fail to help students because they do not provide useful information which students need to make better career choices.

Researchers from local and international sources have identified technology as an essential component of educational systems. The combination of artificial intelligence and chatbot systems together with digital learning platforms represents a major breakthrough for enhancing educational environments. The systems create personalized learning pathways which deliver instant assessments and provide students with continuous assistance throughout their learning process. The studies demonstrate that interactive learning tools together with interactive methods which include gamification and microlearning can boost student engagement and motivation while improving their learning results in all educational settings.

The research studies from local areas and international territories show different patterns of technological growth and deployment. The current state of TVET programs is the main focus of local studies which examine existing problems and suggest improvements through better curriculum design and facility development and teaching method enhancements. The research studies present new solutions through their introduction of gamified systems and AI-supported tools but these solutions still have limited functionality because they have not achieved widespread use. The foreign research literature shows better technology applications which include AI-driven chatbots and natural language processing systems and intelligent learning platforms. The systems create educational support through their capacity to deliver instant help and mimic human conversation and create customized learning experiences.

Foreign studies concentrate more on educational systems through which AI technologies become integrated into their educational framework which lets them use AI for language instruction and vocational education and academic guidance. The research addresses multiple important topics which include data protection standards and ethical principles and system security measures and necessary teacher development for technology implementation. The local studies examine organizational and infrastructure problems at a higher rate than they investigate the ethical and technical difficulties which emerge from using advanced systems.

The research gaps found in this study emerge from the combined analysis of existing literature and research studies. First, existing systems which use artificial intelligence or chatbot technology for educational purposes and employment guidance remain underdeveloped and lack practical application in the local area. Second, the research studies show the value of competency-based training which aligns with industry needs but intelligent technologies remain underutilized for process improvement. Third, there is insufficient focus on creating interactive, user-friendly, and accessible systems that cater to diverse learners and improve engagement. Fourth, local research does not provide sufficient study of system security issues and ethical data usage and implementation methods. The systems currently available need to become more extensive by integrating career guidance with skills assessment and learning support into one unified platform.

The current research benefits from these results which demonstrate their value. The research shows that technical-vocational education requires improvements which need new technology to better prepare students for their future careers. The differences between two systems present an opportunity to transfer advanced technology solutions from international markets into the local environment. The researchers use existing gaps to create their proposed system which will enable them to design a solution that is both efficient and user-friendly.

The synthesis demonstrates that educational and training system improvements have made substantial progress but require development of advanced integrated systems which use modern technological solutions. The current research aims to create a better system which improves student learning outcomes and develops their skills while preparing them for current workplace requirements by addressing existing research gaps.

CHAPTER III

METHODOLOGY

Research Design

This study employs the developmental research design in the creation of the proposed system entitled "TESDA-Bot: An Interactive Guide for TechVoc Students to Prepare for National Certificate (NCII) Exams." This approach emphasizes the systematic planning, design, development, testing, and as well as the evaluation of the system to ensure that it is functional and effective in addressing the users needs.

The software development methodology that will be used in this study will be the Agile Model. This model follows an iterative and flexible process that supports continuous improvement and regular feedback. This includes stages of planning, designing, development, testing, and ongoing revisions. Each phase allows adjustments to be made based on the user input and evaluation of the system before proceeding further.

In addition, the descriptive method is utilized to assess the system's performance through the evaluation of selected users. A survey questionnaire is administered to measure the system in terms of usability, reliability, efficiency, and functionality.

Sampling Technique

In this study, the researchers will use purposive sampling to choose the sample. Purposive sampling is a technique to choose samples based on specific consideration [61]. The respondents selected are knowledgeable about the system, an actual user of the TESDA-Bot, and belong to the target group, specifically, the Technical-Vocational students who are preparing for the National Certificate (NCII) examinations. This ensures that the data collected is reliable and directly related to the objectives of the study.

Participants of the Study

The participants of this study consist of Tech-Voc (Technical-Vocational–Livelihood) students who are preparing for or have recently taken National Certificate II (NCII) certification exams under the TESDA framework. These include senior high school TVL students enrolled in NCII-aligned programs (Computer Systems Servicing NCII, Bread and Pastry Production NCII, Housekeeping NCII, Automotive Servicing NCII, etc.) and, where applicable, TESDA TVET trainees from accredited technical-vocational institutions who are at the level where they are expected to undergo written and practical competency assessments.

Participants will be selected using purposive sampling to ensure that they have actual exposure to NCII-related competencies and are in the stage of exam preparation or post-exam reflection. Their role in the study will vary: some will act as respondents by answering surveys on their preparedness, learning behavior, and satisfaction with *TESDA-Bot*, while others may serve as interview or focus-group participants to provide deeper insights into how an interactive chatbot-based tool can support their review, practice, and confidence in facing NCII assessments.

Research Locale

This study will be conducted in selected Technical-Vocational–Livelihood (TVL) tracks within a senior high school (or TVET institution) in Metro Manila, National Capital Region, Philippines, where Tech-Voc students are enrolled in NCII-aligned programs (Computer Systems Servicing NCII, Automotive Servicing NCII, Bread and Pastry Production NCII, or similar tracks). The physical setting includes classrooms, computer laboratories, and guidance or training offices where students receive TVL instruction and exam-preparation support, and where they will interact with or evaluate TESDA-Bot as an interactive review guide.

In addition, a virtual component will be employed through an online or school-hosted platform where the TESDA-Bot chatbot is accessed, allowing students to use the tool during or after class sessions. This mixed physical-virtual locale is chosen because it reflects the real-world environment where Tech-Voc learners prepare for NCII assessments, making it suitable for observing how an interactive guide can support their review, practice, and confidence before the actual competency exams.

Research Instrument

The proponents used a data collection instrument to gather information. The researchers will use a survey questionnaire as the primary research instrument to evaluate the proposed system. A Google Form survey will be utilized to collect data, featuring questions that employ a Likert scale to measure respondents' attitudes and perceptions. The survey questionnaire is distributed to selected participants after they use the system.

The questionnaires will use a 5-point Likert scale in interpreting the respondents' level of agreement with different aspects of the system. This includes measuring their perceptions of the system's functionality, usability, reliability, and as well as efficiency. Each scale point allows the respondents to express their degree of agreement, ranging from strong disagreement to strong agreement, which will provide a more detailed and accurate assessment of the system's overall performance and effectiveness.

Data Gathering Procedure

The data gathering procedure for TESDA-Bot: An Interactive Guide for TechVoc Students to Prepare for National Certificate (NC II) Exams will be conducted in a systematic and ethical manner to ensure that the information collected was relevant, accurate, and reliable. The researchers first identified the target respondents, which included TechVoc students, teachers, and other individuals who could provide meaningful input regarding NC II exam preparation and the support features needed in the proposed system. After obtaining approval from the concerned authorities, the researchers explained the purpose of the study, the nature of the participants' involvement, and the confidentiality of their responses before securing their consent.

Data were gathered through surveys, interviews, and evaluation forms to determine the students' preparation needs, learning difficulties, and preferred forms of interactive support. The information collected served as the basis for identifying the essential features of the system, such as review assistance, practice questions, exam guidance, and progress tracking. After the prototype was developed, selected respondents evaluated the system in terms of usability, functionality, and effectiveness. Their feedback was then analyzed and used to improve and refine the system before final implementation. Throughout the process, the researchers maintained confidentiality and used the data solely for academic purposes.

System Development Process

The proposed TESDA-Bot: An Interactive Guide for TechVoc Students to Prepare for National Certificate (NC II) Exams will follow the Agile SDLC methodology, which emphasizes iterative development, continuous testing, and gradual improvement. The process begins with requirements gathering, where the needs of TechVoc students and other stakeholders are identified. This is followed by system design, development, testing, and deployment. Each phase is completed in short cycles or sprints, allowing the researchers to incorporate feedback and make necessary revisions throughout the development process. Through this approach, the system is expected to become a user-friendly and effective interactive guide for NC II exam preparation.

System Architecture

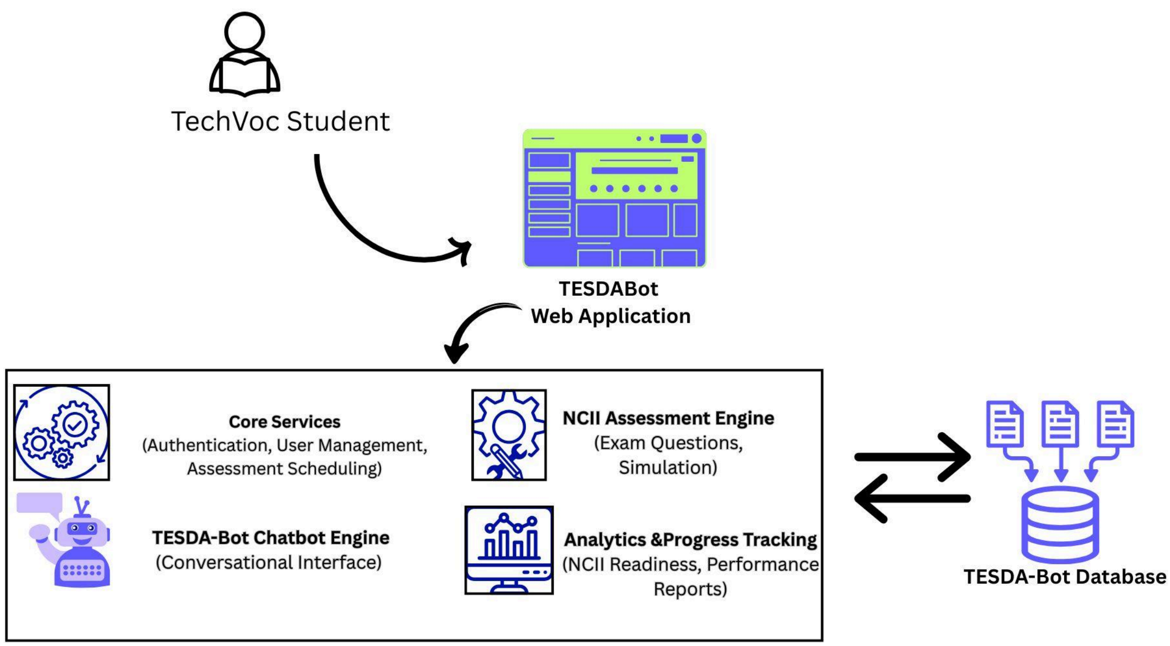


Figure 2. TESDA-Bot System Architecture

The TESDABot system architecture shows a web-based platform where a TechVoc Student interacts with the TESDABot Web Application as the main entry point for learning and assessment activities. The application connects to core services that handle authentication, user management, and assessment scheduling, while the chatbot engine provides a conversational interface for questions and guidance. It also includes an NCII Assessment Engine for generating exam questions and simulations, and an Analytics & Progress Tracking component that monitors NCII readiness and performance reports. All of these functions rely on the TESDABot Database, which stores and retrieves user data, assessment records, chatbot interactions, and progress information, making the system a complete and integrated support tool for TechVoc students preparing for NCII certification.

Data Analysis

The data analysis for TESDA-Bot: An Interactive Guide for TechVoc Students to Prepare for National Certificate (NC II) Exams explains how the collected responses are processed, interpreted, and presented to determine the effectiveness of the proposed system. After the data are gathered, the researchers will organize and encode the responses to prepare them for analysis. The results from surveys, interviews, and system evaluations will be summarized using descriptive statistics such as frequency, percentage, weighted mean, and standard deviation, depending on the type of information collected.

The analysis will focus on the system's usability, functionality, accessibility, and usefulness in helping TechVoc students prepare for NC II examinations. For rating-scale responses, the weighted mean will be used to determine the overall evaluation of the system. The findings will be presented through tables and narrative interpretation to make the results clear and understandable. Through this process, the researchers will identify the strengths and weaknesses of the proposed system and determine whether it effectively supports students in their exam preparation and learning needs.

Data Flow Diagram

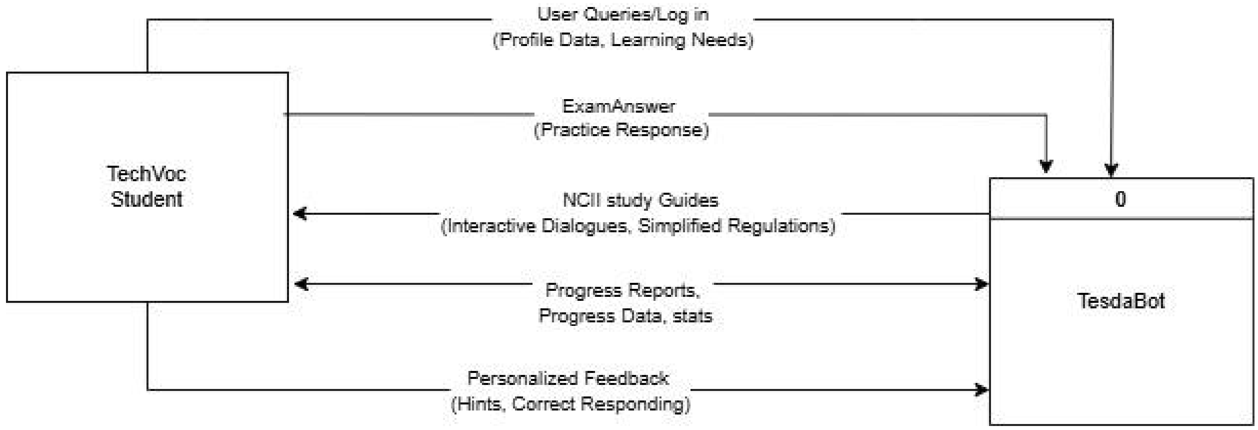


Figure 3. Data Flow Diagram Level 0

The TESDA-Bot Data Flow Diagram illustrates a continuous exchange of information between the TechVoc Student and the TesdaBot system. The process begins when the student provides login credentials and specific learning requirements, which the system uses to deliver tailored NCII study guides and interactive materials. As the student engages with the bot by submitting practice exam answers, the system processes these inputs to return personalized feedback, hints, and detailed progress reports. This circular flow ensures that the student receives a customized learning experience based on their performance and real-time data processing within the bot.

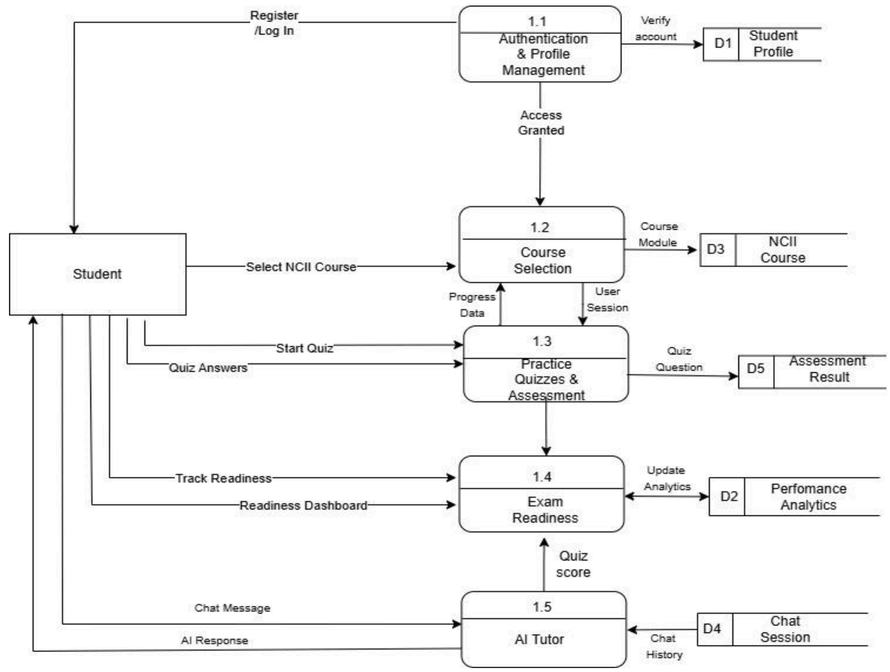


Figure 4. Data Flow Diagram Level 1

The TESDA-Bot Data Flow Level 1 system establishes a systematic path which progresses from user identification to academic assistance based on data. The process starts at Authentication & Profile Management (1.1) because it verifies user credentials through the Student Profile (D1) system to create a safe personalized environment. The Course Selection (1.2) process becomes the access point for NCII Course (D3) content after the system verifies the student's identity because it provides them with study material that corresponds with their academic program. The student uses Practice Quizzes & Assessment (1.3) which causes the system to switch from delivering content to measuring student results when it stores their initial assessment scores in the Assessment Result (D5) database. The Exam Readiness (1.4) process uses the collected performance data to generate Performance Analytics (D2) which converts raw quiz scores into usable information that appears on the student readiness dashboard. The complete process uses AI Tutor (1.5) which utilizes data from Chat Session (D4) archives to deliver precise educational assistance while maintaining ongoing student technical support through flexible data-driven dialogue.

System Flow Chart

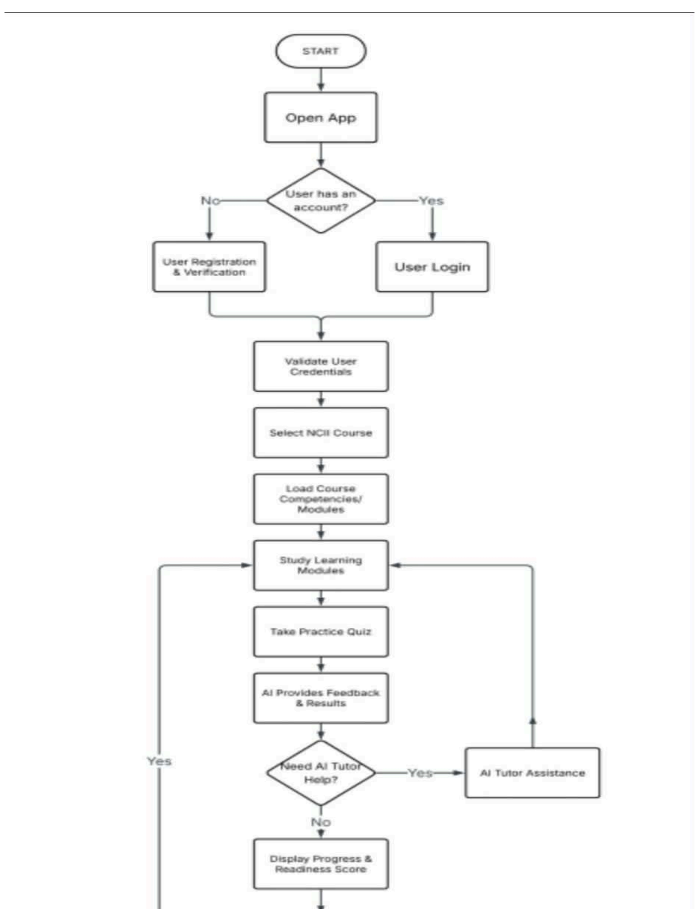


Figure 5. TESDA-Bot System Flow Chart

The TESDA-Bot system flowchart outlines a guided learning process that begins when a user opens the app. Users without an account undergo registration, while those with an account proceed to login, both paths leading to credential validation. After successful validation, users select an NCII course, access related competencies and study modules, and take a practice quiz with AI-generated feedback. Following the quiz, the system prompts the user for AI tutor help; if requested, users receive assistance before returning to the modules. If not needed, progress and readiness scores are displayed. The user is then asked if they wish to continue learning, which can lead back to the study stage or to logging out, concluding the session.

Use Case Diagram

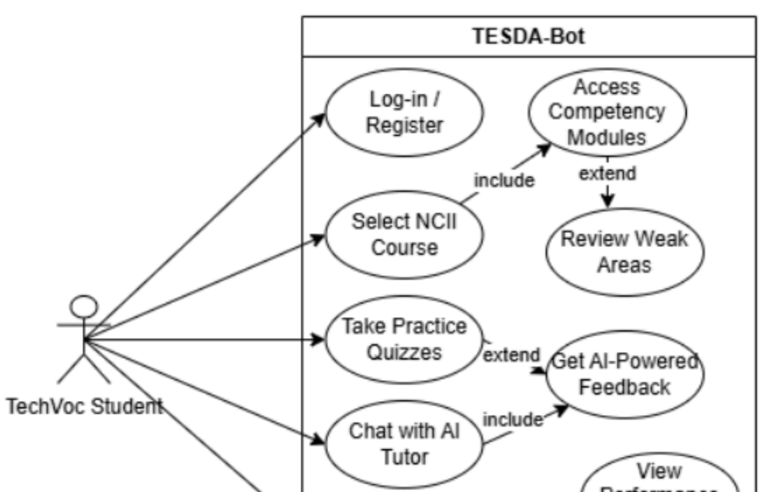
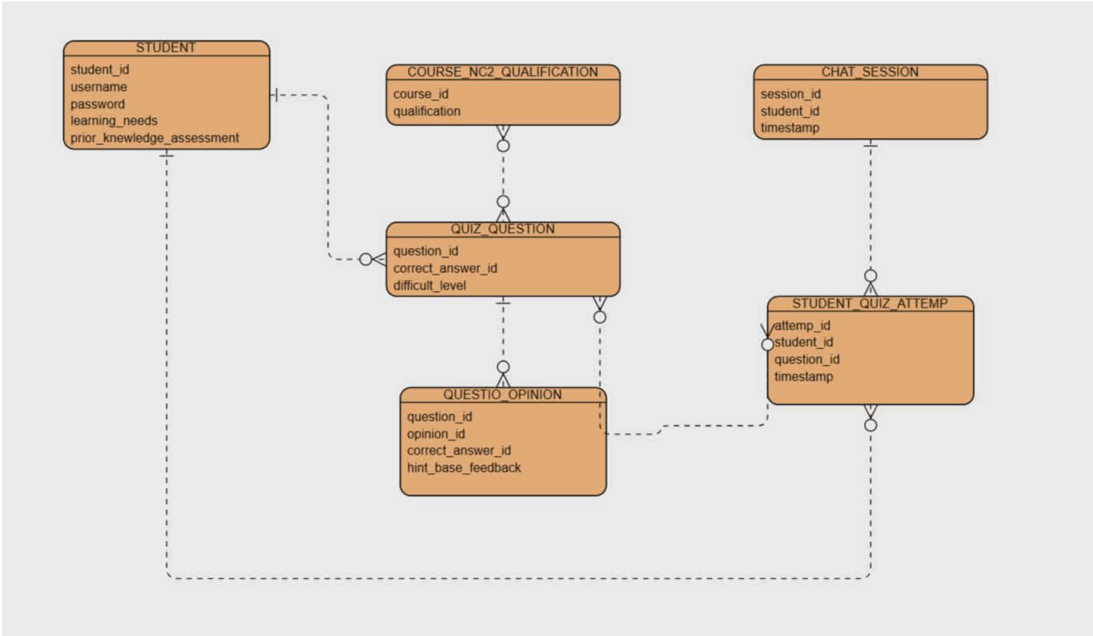


Figure 6. Use Case Diagram

The TechVoc Student accesses the TESDA-Bot by logging in or registering, then selects an NCII course to access competency modules. The system helps identify weak areas for focused study and offers practice quizzes with AI-powered feedback on performance. Students can chat with an AI tutor for guidance and track their exam readiness through progress estimates and performance analytics, enhancing understanding of their strengths and weaknesses.

Entity Relationship Diagram



The ERD for TESDA-Bot depicts an innovative and unique ecosystem aimed at facilitating the preparation of Technical-Vocational students for NCII exams. In terms of system design, the architecture starts with the User entity, which distinguishes between two users based on their roles. The Student entity has an immediate association with the Courses entity since each student is enrolled in courses that prepare them for their chosen trade or qualifications

The core of the learning process lies with the Assessment entity because this is what makes the exam preparation possible. An assessment is made up of several Questions, which include the corresponding Choices to make a proper assessment test. Through interactions with the bot, students' User Responses can be tracked to give the user an insight into their progress and results in the form of a new entity – the Result.

Ethical Considerations

In an IT research or capstone project, ethical considerations play a significant role in ensuring that the project is conducted responsibly and aligns with professional and societal standards. Here are key ethical considerations to keep in mind: (select only which is applicable on your project)

1. Data Privacy and Confidentiality

- **Sensitive Information Protection**: Researchers must ensure that any personal or sensitive data (e.g., health, financial, or identifiable information) is protected and handled with strict confidentiality.

- **Data Anonymization**: If the research involves user data, it's crucial to anonymize it to prevent identification of individuals.

- **Compliance with Data Protection Laws**: Adherence to laws such as GDPR (General Data Protection Regulation) or HIPAA (Health Insurance Portability and Accountability Act) is essential when handling sensitive data.

2. Informed Consent

- **Participants' Knowledge**: When involving human subjects in your research (e.g., surveys, testing), it is important to obtain their informed consent. Participants should be fully aware of what the study involves and any risks.

- **Voluntary Participation**: Participation should be voluntary, without any form of coercion or undue influence, and subjects should be allowed to withdraw at any time.

3. Intellectual Property and Plagiarism

- **Original Work**: Capstone projects should avoid plagiarism, ensuring proper citation of sources and acknowledgment of previous research.

- **Copyright and Licensing**: Use of third-party software, libraries, or tools should respect copyright and licensing agreements.

4. Social Impact and Responsibility

- **Impact on Society**: The potential societal effects of the project should be evaluated. Avoid projects that could have negative consequences, such as violating human rights or enabling harmful practices.

- **Sustainability and Inclusivity**: Consider environmental sustainability and inclusivity, ensuring that technology developed benefits a broad range of users without discrimination.

5. Cybersecurity and Protection from Harm

- **Ensuring Security**: If the project involves software development, ethical consideration must include building secure and resilient systems to protect users from cyber threats.

- **Avoiding Harm**: Projects should avoid causing harm to individuals or systems, whether through negligence, vulnerabilities, or malicious design.

6. Bias and Fairness

- **Algorithmic Bias**: If the project involves algorithms, ensure that they are fair and unbiased. Biased systems can lead to discriminatory outcomes, so fairness should be a priority.

- **Transparency**: Be transparent about how data is used and analyzed to prevent hidden biases from affecting results.

7. Accountability and Integrity

- **Accuracy in Reporting**: Researchers should report findings honestly, without fabricating, falsifying, or misrepresenting data or results.

- **Responsibility for Actions**: Take responsibility for the ethical conduct of your research, including acknowledging any limitations or weaknesses in the study.

8. Dual-Use Technology

- **Potential for Misuse**: Consider if the technology or research developed could be used for both beneficial and harmful purposes, and take steps to mitigate any risks of misuse.

9. Collaboration Ethics

- **Equal Credit**: In collaborative projects, ensure that all contributors receive appropriate credit, and no one's contributions are overlooked.

- **Conflict of Interest**: Disclose any conflicts of interest, such as personal, financial, or professional connections that could affect the research outcomes.

These ethical considerations ensure that an IT capstone project or research is conducted responsibly, safeguarding the interests of individuals, society, and the broader tech community.

REFERENCES

[1] Asis, L. C. (2020b). Employability of Senior High School Graduates under TECHVOC Track with National Certification in Graphics and Animation from TESDA. *The Educational Review USA*, *4*(12), 219–224.

[2] Tiyanes, T. M. M., & Ponsades, O. P. (2024). EMPLOYMENT JOURNEY OF TECHNICAL TEACHERS: STORIES TO TELL. *European Journal of Social Sciences Studies*, *10*(3).

[3] Alinea, Jess Mark & Garcia, Christian & Noblefranca, Sheila & Lagado, Suzette. (2024). Employability Skills of Senior High School Tech-Voc Students: A Measure of Industry Preparedness.

[4] Cao-Wat, K. C., Invierno, B., & Manuel, J. (2026). BRIDGING LOCAL STANDARDS AND GLOBAL COMPETENCIES: ASSESSING CONTACT CENTER SKILLS BASED ON TESDA NC II FRAMEWORK. *ShodhGyan-NU Journal of Literature and Culture Studies*, *4*(1), 61–67.

[5] Wacas, Ronald. (2020). Effectiveness of E-Learning Technology on The Performance of the CSS NC II Trainees (Computer Systems Servicing) of Tesda Accredited Training Centers in TESDA-Kalinga. International Journal of English Literature and Social Sciences. 5. 2398-2405. 10.22161/ijels.56.77.

[6] R. Monteverde, I. Mejia, L. Dasas, S. Bazer, C. M. Garcia, and C. Aquino, “Assessing the current and future middle skills in the Philippines: Inputs for Policy agenda,” Aug. 2025. doi: 10.62986/dp2025.22.

[7] C. A. L. Jaca, F. B. Javines Jr., and JACA & JAVINES JR., “Hybrid Collaborative Model of Technical-Vocational-Livelihood (TVL) track in senior high school in the Philippines,” journal-article, 2022.
[8] M. R. Caig and Pagbilao National High School, “Cultivating Lifelong Learners through Revitalized Technology Livelihood Education and Technical Vocational Livelihood Programs for Public Secondary Schools,” journal-article, 2022.
[9] E. S. Dungca, “Work Readiness of Industrial Arts Graduates of Technical Vocational: A Sequential Explanatory Study,” *GEAPHEI Research Publication*, Nov. 2025, [Online]. Available: https://geaphei.org/viewArticle.php?id=18
[10] M. B. BULANADI, “EXTENT OF IMPLEMENTATION OF COMPETENCY-BASED PROGRAM OF TECHNICAL VOCATIONAL EDUCATION IN a PUBLIC HIGH SCHOOL: BASIS FOR SKILLS DEVELOPMENT PERFORMANCE,” 2014.
[11] S. S. Lorenzo and J. M. Claravall, “LEVEL OF COMPETENCY OF SENIOR HIGH SCHOOL STUDENTS UNDER STRENGTHENED TECHNICAL VOCATIONAL EDUCATION PROGRAM (STVEP) AND FACTORS AFFECTING IT: a BASIS FOR ENHANCEMENT PROGRAM,” *Open MIND*, Mar. 2026, doi: 10.5281/zenodo.18908009.
[12] F. G. Ramos PhD, “An Evaluation of the Technical Vocational Livelihood Track in Public Senior High Schools in the Division of Batangas:

Basis for an enhancement program," 2021. doi: 10.6007/IJARPED/v10-i2/10269.

[13] A. H. Dumaplin and Sorsogon State University, "Factors that influence technical vocational students' choice of specialization in relation to their academic performance in Deped Division of Sorsogon," 2024.

[14] J. T. Maestrado and J. D. Bucar, "Career Guidance Advocacy Program and the Grade 10 Completers' Career Choice," *International Journal of Multidisciplinary Applied Business and Education Research*, vol. 5, no. 2, pp. 513–527, Feb. 2024, doi: 10.11594/ijmaber.05.02.12.

[15] R. I. Navasca, M. G. Navasca, and N. B. C. Tugelida, "Technology and Livelihood Education in Northern Philippines: A Comprehensive case study of DMMMSU- NLUC," Mar. 31, 2025.

[16] C. B. Chua, "Rationalizing TVET provision and support for enhanced public‐private complementarity," Ateneo and EDCOM‐II Research Fellowship Paper no. 14, Dec. 2025.

[17] K. C. Cao‐wat, B. O. Invierno, and J. B. Manuel, "Bridging local standards and global competencies: Assessing contact center skills based on TESDA NC II framework," *ShodhGyan‐NU: J. Lit. Cult. Stud.*, vol. 4, no. 1, pp. 61–67, 2026.

[18] MR. M. R. Gualiza, "Teachers' qualification, facilities and competencies of the Grade 7 students in Technology and Livelihood Education," *Turk. Online J. Design Art Commun. (TOJDAC)*, vol. 7, Special Issue – DCMSPCL, pp. 106–114, 2024.

[19] A. C. Orbeta and J. P. P. Corpus, “The national system of technical vocational education and training,” *Philippine Inst. Dev. Stud. (PIDS) Discuss. Pap. Ser.*, no. 1607, 2016.

[20] J. P. Antonio, “Utilization, challenges, and opportunities of AI‑driven English language development software of language teachers,” *Int. Online J. Educ. Res. (IOER‑IMRJ)*, 2024.

[21] B. J. Dr. Pasion, “Innovation and Digitalization Strategy in Technical Education: Assessment of an ergonomic universal working Table for optimized instructional performance,” *Questa Soft*, 2025.

[22] M. A. T. Mercado, “CSS QUEST: GAMIFYING COMPUTER SYSTEM SERVICING MODULE USING GAMIFIED PROGRESS TRACKING AND INTERACTIVE STORYTELLING ALGORITHM FOR GRADE 12 STUDENTS IN CALAMBA CITY,” International Journal of Advanced Research in Computer Science, vol. 14, no. 03, pp. 60–79, Jun. 2023, doi: 10.26483/ijarcs.v14i3.6996.

[23] M. C. A. Elli and J. D. Ricafort, “Competencies of Grade VI teachers in Technology and Livelihood Education (TLE).”

[24] I. Marcojos-Canon, “Linking the DOTS of STEM instruction in private higher education institution,” Jul. 2018.

[25] A. Ferrer and Polytechnic College of Botolan, “Status of implementation of Technical-Vocational-Livelihood (TVL) track in secondary schools in the District of Botolan,” 2021.

[26] J. C. Nicodemus, “QUALITATIVE EXPLORATION OF THE TRANSITION EXPERIENCES OF THE SENIOR HIGH MODEL SCHOOL GRADUATE IN REGION IV-CALABARZON,” 2021.

[27] E. A. Cundangan, “THE RELATIONSHIP OF THE SENIOR HIGH SCHOOL TRACKS AND STRANDS, AND THE CURRICULUM EXITS OF THE GRADUATES,” journal-article, Oct. 2023.

[28] M. J. B. Mamayabay and M. R. T. Quilestino, “MOTIVATIONS AND CHALLENGES IN SPECIALIZING TECHNOLOGY AND LIVELIHOOD EDUCATION: A PHENOMENOLOGICAL INQUIRY,” *European Journal of Education Studies*, vol. 11, no. 2, Feb. 2024, doi: 10.46827/ejes.v11i2.5205.

[29] F. I. V. CONSOLACION Main Author, B. J. A. COSOLACION, and E. T. MAAGAD, “COMPETITIVENESS OF TESDA NC II CERTIFICATION IN TOURISM SECTOR,” ZAIN Publications, research paper, Mar. 2025. doi: 10.47760/cognizance.2025.v05i07.003.

[30] Yoshida, Yoshimitsu & Guzman, Rodel. (2025). Communication Skills and Work Readiness of Senior High School Technical-Vocational-Livelihood Track Students. Journal of Interdisciplinary Perspectives. 3. 10.69569/jip.2025.152.

[31] Fitria, T. N., Simbolon, N. E., & Afdaleni, A. (2023). Chatbots as online chat conversation in the education sector. *International Journal of Computer and Information System (IJCIS)*, *4*(3), 93–104.

[32] W. Wiboolyasarin, “AI-driven chatbots in second language education,” *Comput. Assist. Lang. Learn.*, 2025.

[33] C. Fernandez, M. Balijon, and I. Ancho, “Career preferences of Filipino senior high school students,” *CTU Journal of Innovation and Sustainable Development*, vol. 15, no. 3, pp. 81–91, Nov. 2023, doi: 10.22144/ctujoisd.2023.053.

[34] C. D. Aguilar, “Integrating artificial intelligence in secondary English instruction: Regulatory practices and issues encountered,” *Int. J. Stud. Engl. Lang. Literat. (INJOSEDU)*, vol. 13, no. 10, pp. 2441–2458, Oct. 2025.

[35] J. D. Cruz, M. A. L. Santos, and R. G. Manligoy, “AI-based career path recommender system for senior high school students using multi-label classification,” *Digital Systems and Management (DSM)*, vol. 2, no. 4, pp. 102–109, 2025.

[36] O. Zahour, E. H. Benlahmar, A. Eddaoui, H. Ouchra, and O. Hourrane, "A system for educational and vocational guidance in Morocco: Chatbot E-Orientation," Procedia Computer Science, vol. 176, pp. 2470-2477, 2020

[37] M. A. Al-Ghamdi, "An Intelligent Career Counseling System for Saudi Students Based on Career Ontologies," in Proc. 2018 Int. Conf. Smart Commun. Netw. Technol. (SaCoNeT), 2018, pp. 1-6

[38] A. Tlili, B. Chang, T. He, F. Huang, and Z. Liu, "AMChat: An AI-based Minecraft Chatbot for Assisting Student Game Design," in Blended

Learning: Education in a Smart Learning Environment, W. W. K. Ma et al., Eds. Cham, Switzerland: Springer, 2019, pp. 565-574

[39] T. K. F. Chiu and B. L. Moorhouse, "Teacher support and student motivation to learn with Artificial Intelligence (chatbot)," Interact. Learn. Environ., 2023

[40] M. A. Al-Ghamdi, "Intelligent Career Counseling Chatbot for Saudi Students Using Career Ontologies," in Advances in Intelligent Systems and Computing, A. Abraham et al., Eds. Cham, Switzerland: Springer, 2019, pp. 301-310

[41] T. B. Prassetiyo et al., "Intelligent Career Counseling Chatbot for Saudi Students Using Career Ontologies," in Advances in Intelligent Systems and Computing, A. Abraham et al., Eds. Cham, Switzerland: Springer, 2019, pp. 301-310

[42] W. F. M. Heijnen and A. A. Verbraeck, "The design of a chatbot for supporting an interactive e-learning environment for introductory programming courses," Computers & Education, vol. 139, pp. 82-98, Oct. 2019.

[43] M. A. Al-Ghamdi et al., "Intelligent Career Counseling System for Saudi Students Using Career Ontologies," in Proc. 2020 IEEE Int. Conf. Innov. Intell. Syst. Appl. (INISTA), 2020, pp. 1-6

[44] J. Walton, R. L. Walton, and A. Lodge, "Qualitative exploration of digital chatbot use in medical education: A pilot study," Digital Health, vol. 7, 2021.

[45] C. Kooli, "Chatbots in Education and Research: A Critical Examination of Ethical Implications and Solutions," Sustainability, vol. 15, no. 7, Art. no. 5614, Apr. 2023

[46] S. Bourdages, “Utilisation, défis et perspectives de l’utilisation de logiciels de développement de la langue anglaise soutenus par l’intelligence artificielle par les enseignants de langue,” *J. des technologies de l’apprentissage* (JTL), vol. 19, no. 4, 2025.

[47] V. Pham and T. P. T. Thuy, “Integrating artificial intelligence in undergraduate tourism education to meet international standards,” *DS J. Multidiscip.*, vol. 2, no. 4, pp. 17–24, 2025

[48] M. A. L. Santos and J. P. Capuno, “Navigating futures: Examining how the addition of senior high school tracks influenced postsecondary aspirations among education majors in the Philippines,” *Asia Dev. Rev.*, 2025.

[49] M. R. Gualiza, “Teachers’ qualification, facilities and competencies of the Grade 7 students in Technology and Livelihood Education,” *Turk. Online J. Design Art Commun. (TOJDAC)*, vol. 7, no. DCMSPCL, pp. 106–114, 2024.

[50] A. B. Researcher, C. D. Educator, and E. F. Scholar, “Title of the article,” *Adv. Multiling. Lang. Educ. Res. (AMLER)*, vol. 1, no. 1, pp. 1–10, 2026.

[51] S. S. Alotaibi and A. M. Alotaibi, "Developing an ontology-based intelligent career counseling system for Saudi Arabian students," in Proc. 2018 IEEE Glob. Eng. Educ. Conf. (EDUCON), 2018, pp. 811-816

[52] M. A. Al-Ghamdi et al., "An Intelligent Career Counseling System for Saudi Students Based on Career Ontologies," in Proc. 2018 Int. Conf. Smart Commun. Netw. Technol. (SaCoNeT), 2018, pp. 1-6

[53] T. K. F. Chiu, "Digital support for student engagement in blended learning based on Self-Determination Theory," J. Res. Innov. Teach. Learn., 2021 , vol. 15, no. 1, pp.

[54] A. Deveci Topal, C. D. Eren, and A. Kolburan Geçer, "Chatbot application in a 5th grade science course," Education and Information Technologies, vol. 27, no. 2, pp. 2021-2042, 2022.

[55]A. Tlili, O. Shehata, M. A. Adarkar, and D. Liu, "Is Metaverse in education living up to the hype? A critical review of educational Metaverse and future research agenda," in Emerging Technologies for Education, L. H. Wong et al., Eds. Cham, Switzerland: Springer, 2019, pp. 394-403

[56] T. B. Prassetiyo and A. Author, "Career guidance system development using AI and multiple intelligence assessment," J. Inf. Syst. Eng., vol. 22, no. 1, pp. 101-145, 2022.

[57] T. K. F. Chiu, "The effects of using a chat bot on student learning outcomes and goal attainment in problem-based learning," J. Res. Innov. Teach. Learn., vol. 14, no. 1, pp. 104-116, 2021.

[58] H. L. Chen, G. V. Widarso, and H. Sutrisno, "A ChatBot for Learning Chinese: Learning Achievement and Technology Acceptance," J. Educ. Comput. Res., vol. 58, no. 6, pp. 1123-1149, 2020.

[59]T. B. Prassetiyo and A. Author, "Career Guidance System Development Using AI and Multiple Intelligence Assessment," J. Inf. Syst. Eng., vol. 7, no. 25, pp. 10114509, 2022.

[60]A. A. Alghamdi, B. B. Alghamdi, and C. C. Alghamdi, "NextGen Career Advisor: Hybrid AI System for Personalized Student Guidance Using Multimodal Data Fusion," in Proc. 2024 IEEE Int. Conf. Educ. Technol. (ICETC), 2024, pp. 1-8